Ideation in Software Design

Module 3 (09)

216U01E701

Software Design Thiniking

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The Design Thinking Roadmap: From Creative Spark to Feasible Solution

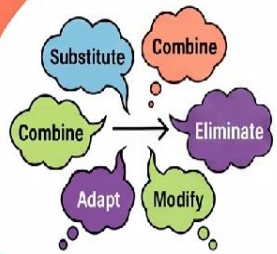
A step-by-step guide to generating, visualizing, and prioritizing innovative design alternatives

IDEATION—EXPANDING THE SOLUTION SPACE



Divergent Thinking & Mind Mapping

Multiply options by using mind maps to visualize non-linear connections between core themes.



Structured Brainstorming & SCAMPER

Use "trigger questions" and SCAMPER techniques to modify existing ideas into new alternatives.



The Power of "Building On"

Follow the "Yes, and..." rule to evolve individual ideas into robust, shered concepts.

MAPPING—VISUALIZING THE EXPERIENCE

User Journey Mapping

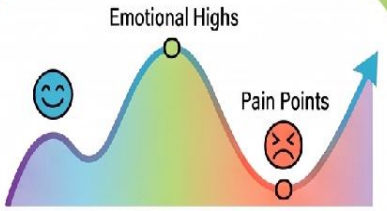
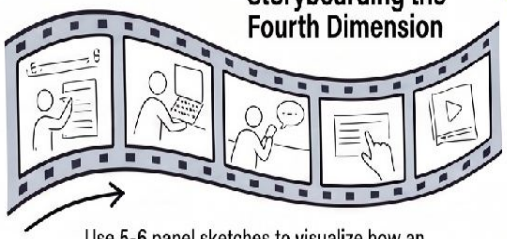


Chart the user's emotional highs and lows to identify "pain points" and innovation opportunities.

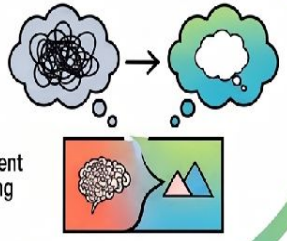
Storyboarding the Fourth Dimension



Use 5-6 panel sketches to visualize how an interaction unfolds over time.

Visualization as Risk Mitigation

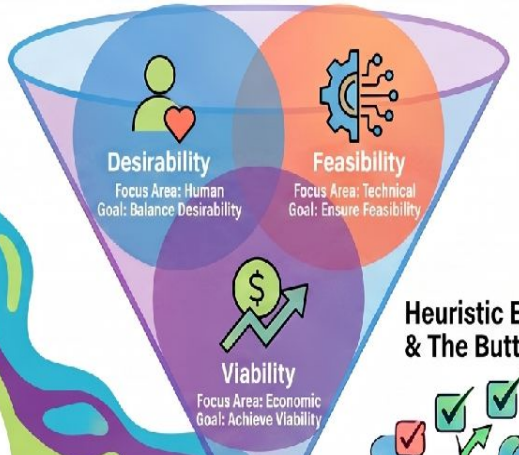
Use imagery to align mental models and prevent team misalignment during complex execution.



CONVERGENCE—PRIORITIZATION AND FEASIBILITY

The Three Constraints of Innovation

Balance human desirability, technical feasibility, and economic viability to ensure long-term success.



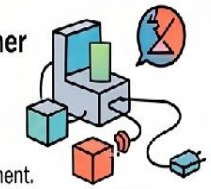
Heuristic Evaluation & The Butterfly Test



Use expert rules and group "voting" to converge on the most promising design nodes.

Fail Early to Succeed Sooner

Use low-fidelity prototypes to test key assumptions before investing in high-cost development.





Feasible Solution

Gemini Notebook

Module 3: Ideation in Software Design

- Creativity techniques: brainstorming, SCAMPER, mind mapping
- Generating multiple design alternatives
- Storyboarding and user journey mapping
- Design heuristics for software solutions
- Evaluating idea feasibility and prioritization

Ideation

Ideation is the process of generating, exploring, and developing multiple possible solutions to a clearly identified user problem.

In **Software Design**, ideation happens after understanding the user and defining the problem.

- **Ideation = Generating many possible software solutions before selecting the best one.**
- The goal is **not to immediately find the perfect solution**, but to explore a wide range of possibilities.

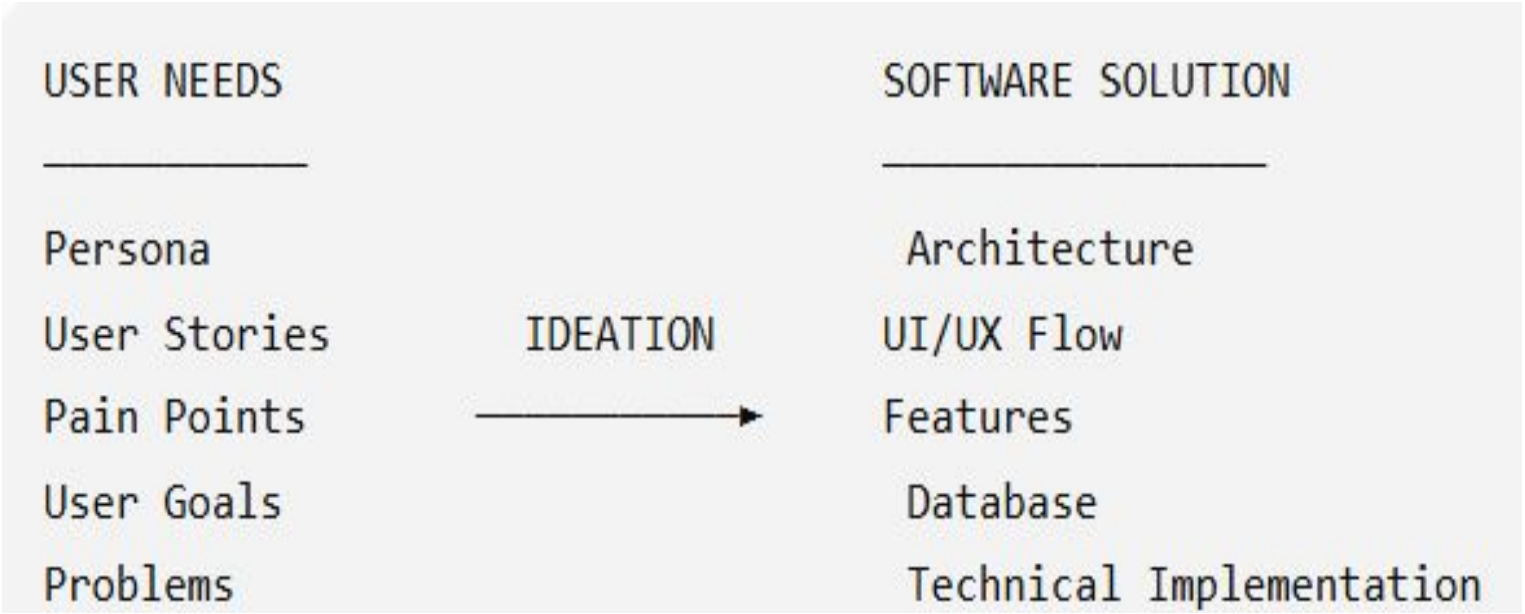
Ideation

- Bridges user needs (user stories, personas, pain points) and concrete software concepts (architecture, UI/UX flows, feature sets, technical implementations)

- Transform abstract user problems into high-value, technically sound software concepts before costly development begins.

Bridges User Needs and Software Concepts:

Ideation acts as a bridge between what users need and what developers actually build.



Creativity techniques: brainstorming, SCAMPER, mind mapping

- **Brainstorming is a group creativity technique in which participants generate a large number of ideas for a problem without judging or criticizing the ideas during the initial generation phase.**

Main Objective:

- **Generate quantity first, evaluate quality later.**

.

Brainstorming :Key Principles of Brainstorming

1. Defer Judgment

Do not immediately say:

"That idea won't work."

Instead, record it and evaluate it later.

2. Encourage Quantity

The team should generate **many ideas** rather than trying to find one perfect solution immediately.

3. Encourage Wild Ideas

Unusual ideas can lead to practical innovations.

Example:

"What if students could talk to the college ERP like ChatGPT?"

Continue..

4. Build on Others' Ideas

One person's idea can become the starting point for another idea.

Example:

Student A: "Send attendance notifications."

Student B: "Add prediction to tell students if their attendance may fall below 75%.“

5. Keep the Problem Clearly Defined

Brainstorming should focus on a specific **How Might We (HMW)** question.

Example:

How might we help students manage their academic deadlines?

Brainstorming Process

Identify Problem

↓

Define HMW Question

↓

Set Rules

↓

Generate Ideas

↓

Record Ideas

↓

Group Similar Ideas

↓

Evaluate

↓

Select Best Ideas

Activity 1

"How can we improve the college library experience?“

How I can improve the LMS experience?

SCAMPER

SCAMPER is a structured creativity technique used to modify an existing product, service, process, or idea to generate new alternatives.

Instead of asking:

"What new software should we create?"

we ask:

"How can we change the existing software?"

Letter	Meaning
S	Substitute
C	Combine
A	Adapt
M	Modify / Magnify / Minify
P	Put to another use
E	Eliminate
R	Reverse / Rearrange

- **S – Substitute**

Question

What can we replace?

- Think about replacing:
- Technology
- Interface
- Process
- Input
- Output
- User interaction

Software Example

Existing:

College attendance through biometric machine

Substitute:

- Biometric → QR code
- QR code → Face recognition
- Face recognition → Mobile NFC

Another Example

- Traditional customer support:
- **Phone call → AI chatbot**

C – Combine

Question

What can we combine?
Combine two or more functions into one solution.

Software Example
Combine:
College timetable + Attendance + Assignment + Notifications

into:

Student Academic Assistant

2. Example

Combine:

Maps + Food Delivery + Restaurant Reviews

to provide location-based restaurant recommendations.

3. Example

Combine:

Fitness Tracker + Diet Planner + Doctor Consultation

into one health-management platform.

A – Adapt

Question

What existing idea can we adapt from another context?

Look at solutions from other industries.

Example
Banking applications use: **Swipe Left / Swipe Right (**amount transfer)

Adapt this idea for: **Job applications**

Swipe Right → Apply
Swipe Left → Skip

Adapt **Uber's live location tracking** for:

- School bus tracking
- Hospital ambulance tracking
- College shuttle tracking

M – Modify

Question

What can we change, enlarge, reduce, simplify, or redesign?

Example 1

College ERP has a complicated menu with 20 options.

Modify it into:

My Attendance
My Timetable
My Assignments
My Fees
My Exams

Ex 2. Modify an e-commerce search system to provide:
"Show me products under ₹2,000 suitable for a college student."

P – Put to Another Use

Question

- **Can the existing technology or system be used for another purpose?**

Example 1

A **QR code system** originally used for payments can also be used for:

Attendance

- Library book tracking
- Event registration
- Visitor management

Example 2

GPS technology used in:

Food delivery

can also be used for:

College bus tracking.

E – Eliminate

- **Question**

What can we remove?

- Remove unnecessary:

- Features
- Screens
- Steps
- Forms
- Data entry
- Approvals

Example:

Current:
Cart
↓
Address
↓
Delivery Option
↓
Coupon
↓
Payment
↓
Confirmation

Eliminate unnecessary steps and create:

Cart
↓
One-Click Checkout
↓
Payment

R – Reverse / Rearrange

Question

- **What happens if we reverse or change the order?**

Example

Traditional food ordering:

Customer searches restaurant

↓

Selects food

↓

Places order

Reverse the process

AI understands customer's preferences

↓

Suggests food

↓

Customer selects

:

How could we improve WhatsApp for college students?

SCAMPER	Idea
Substitute	**Replace text announcements with short voice/video announcements.**
Combine	**Combine WhatsApp with college timetable.**
Adapt	**Adapt disappearing messages for temporary assignment reminders.**
Modify	**Create a dedicated academic dashboard inside the app.**
Put to Another Use	**Use WhatsApp groups for automated attendance notifications.**
Eliminate	**Remove repetitive messages by introducing automated FAQs.**
Reverse	**Instead of students asking faculty for information, an AI bot proactively provides important updates.**

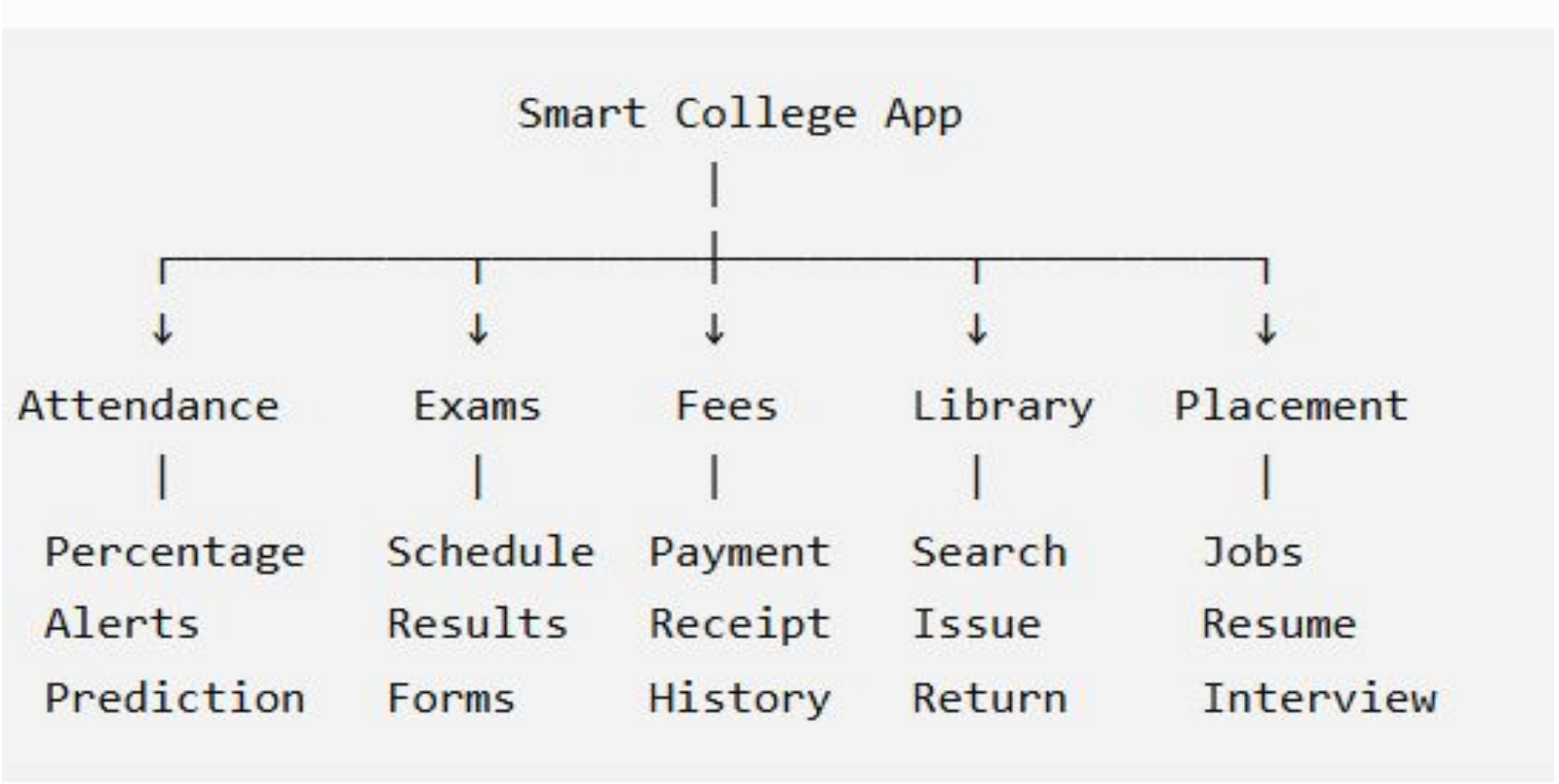
3. Mind Mapping

- **Mind mapping is a visual technique for organizing information and exploring relationships between ideas around a central topic.**

- Instead of generating ideas randomly, students organize them into **branches and sub-branches**.

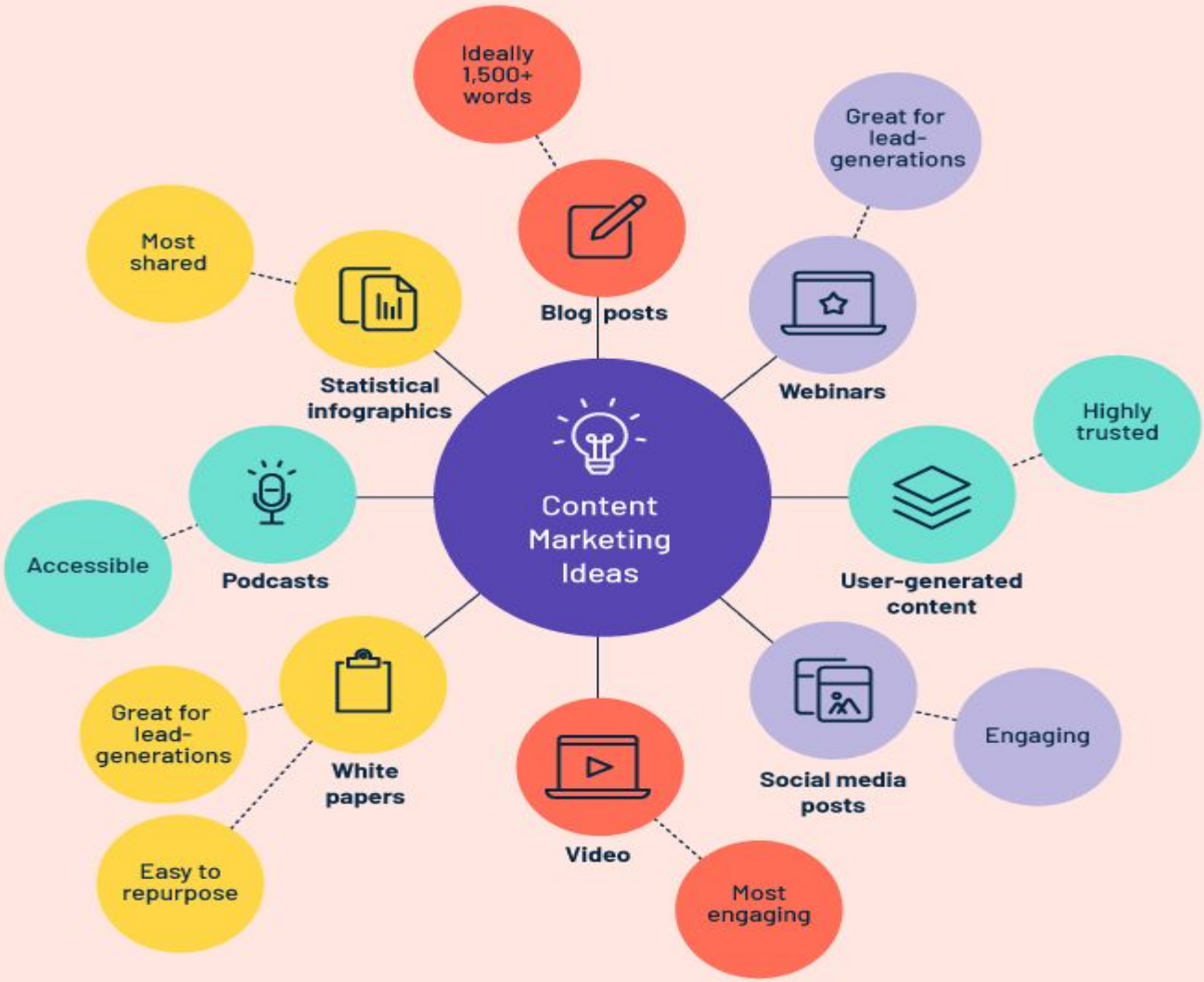
- Mind mapping is a visual brainstorming tool used in design thinking to connect thoughts, structure complex data, and spark creative solutions. It places a core problem or idea in the center and expands outward into categorized branches and sub-branches of words, themes, and user insights

Mind mapping



G2.com

Mind Mapping Example



Mind Mapping vs Brainstorming

Brainstorming	Mind Mapping
Mainly generates ideas	Organizes and connects ideas
Can be individual or group	Usually visual/individual or group
Focuses on quantity	Focuses on relationships
Ideas may initially be unstructured	Ideas are arranged hierarchically
Example: Generate 20 app ideas	Organize app features into categories

Reference:

https://www.g2.com/articles/design-thinking

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Generating multiple design alternatives

How software teams explore, expand, and narrow the space of possible designs.

What We'll Cover

01 What Are Design Alternatives?

02 The Software Design Space

03 Morphological Charts

04 Analogical Thinking

05 Team Ideation Games

06 Navigating the Design Space

What Are Design Alternatives?

When a system can be built in different ways that all satisfy the same requirements — but arrange or implement the same components differently — those different builds are design alternatives.

EXAMPLE — "Let clients fetch and update product data."

- REST API — resource-oriented endpoints, wide tooling support
- GraphQL — single endpoint, client-shaped queries
- gRPC — binary protocol built for fast service-to-service calls

The Software Design Space

The full universe of possible solutions to a problem — every combination of architecture, stack, and interaction pattern that could satisfy the requirements.

- A large design space means many acceptable designs exist to explore.
- A small design space means the problem is tightly constrained.
- Naming the space early gives the team a shared feel for how much room they have.

A Tool for Generating Alternatives

A matrix: each row is a function the design must perform; each column is a different way to achieve it. Pick one means per row to build a complete alternative.

1

List Functions

Every function the system must deliver, one per row.

2

List the Means

Distinct ways to implement each function.

3

Combine

Pick one means per row — that's one candidate design.

4

Prune

Cross out combinations that clash or break. constraints

Example: Designing a Login Feature

Auth Method	Email + Password	**OAuth (Google/GitHub)**	Biometric	Magic Link
Session Handling	**JWT (stateless)**	Server session + cookie	Token in localStorage	
UI Pattern	**Modal overlay**	Full-page form	Inline embedded	
Password Recovery	**Email reset link**	SMS OTP	Security questions	

Highlighted = OAuth + JWT + Modal + Email reset link

Another Example :

Module 3.2_Morphological Chart.docx"

Not Every Combination Is Feasible

Apply interface constraints, security requirements, and common sense to rule out combinations that look valid on paper but break in practice.

✓ FEASIBLE

OAuth + JWT + Modal + Email reset link

Widely supported and keeps auth stateless for. scaling

✗ INFEASIBLE

Biometric + localStorage + Security questions

localStorage tokens are XSS-exposed; security questions are a weak recovery path.

Borrowing Structure From the Familiar

Analogies point out parallels between a new problem and something already familiar — a way to generate ideas without a blank page.

Direct Analogy

The “shopping cart” in e-commerce.

Symbolic Analogy

Files and folders standing in for paper.

Fantasy Analogy

A voice assistant as a companion.

DIVERGENT

expands the design space.

- Remove limits and constraints deliberately
- Widen the store of design ideas and choices

CONVERGENT:

"Which alternatives are actually suitable?"

- Narrow the design space toward the best alternative
- Weigh options against real constraints and goals

The 6-3-5 Method: Brainwriting

Six team members, three ideas each, five silent rotations — keeps every voice contributing without cross-talk.

1

Write

6 engineers each write 3 ideas.

2

Pass

Sheets rotate — no talking, only written notes.

3

Build

Teammates extend the ideas in front of them.

4

Converge

After 5 rotations, the team reviews everything.

05 · TEAM IDEATION

The Gallery Method: Critique Walls

Individual sketches go up on a shared wall — or a Figma board — and become the backdrop for open, iterative critique.

- Each teammate sketches an idea independently, within a time box.
- All drafts are posted together for open questions and critique.
- Each person revises their own draft based on feedback.
- The cycle repeats until the group reaches a saturation plateau.

Know When to Expand or Contract

Expand

When the space feels too small or too few options surface — for a single microservice, a form rule, a caching layer.

Contract

When the space is too large or complex — a platform migration, a multi-service architecture with dozens of variables.

Expanding: Widen Your Options

- Prior Art & Literature — read RFCs and engineering blogs.
- Open-Source Search — reuse existing libraries and frameworks.
- Competitor Benchmarking — try competing products directly.
- Reverse Engineering — inspect how existing APIs achieve a function.

Contracting: Prune Your Options

- Apply Constraints — filter by latency, scale, security, cost.
- Freeze Scope Early — defer nice-to-haves for this iteration.
- Prioritize by Impact — rank options against real requirements.
- “Get Real” Check — apply plain engineering judgment.

FINAL WORD

Resources — time, budget, engineering hours — are always scarce. The goal was never to keep every alternative alive; it was to generate enough of them to make a confident choice.

- Morphological charts
- Team ideation games
- Research & prior art
- Experience and judgment

SCAMPER

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Generate Multiple Alternatives

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Story Boarding

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User Journey Mapping

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Design Heuristics for Software Solutions

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Evaluation Idea Feasibility

- Szmc/x

Prioritization

- Teszmc’m’c,